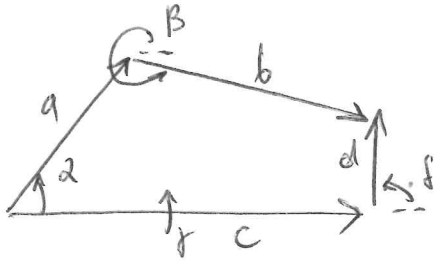


ES 1)

velocity

$$\begin{cases} -a\dot{\alpha}\sin\alpha - b\dot{\beta}\sin\beta = \dot{c}\cos\gamma \\ a\dot{\alpha}\cos\alpha + b\dot{\beta}\cos\beta = \dot{c}\sin\gamma \end{cases} \rightarrow \begin{bmatrix} \cos\gamma & b\sin\beta \\ \sin\gamma & -b\cos\beta \end{bmatrix} \begin{bmatrix} \dot{c} \\ \dot{\beta} \end{bmatrix} = \begin{bmatrix} -a\dot{\alpha}\sin\alpha \\ a\dot{\alpha}\cos\alpha \end{bmatrix}$$

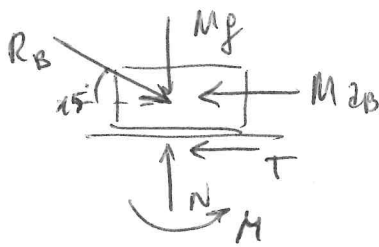
$$\begin{cases} \dot{c} = 0,896 \text{ m/s} = |\vec{v}_B| \\ \dot{\beta} = 0,366 \text{ rad/s} \end{cases}$$

acceleration

$$\begin{cases} -a\ddot{\alpha}\sin\alpha - a\dot{\alpha}^2\cos\alpha - b\ddot{\beta}\sin\beta - b\dot{\beta}^2\cos\beta = \ddot{c}\cos\gamma \\ a\ddot{\alpha}\cos\alpha - a\dot{\alpha}^2\sin\alpha + b\ddot{\beta}\cos\beta - b\dot{\beta}^2\sin\beta = \ddot{c}\sin\gamma \end{cases}$$

$$\rightarrow \begin{bmatrix} \cos\gamma & b\sin\beta \\ \sin\gamma & -b\cos\beta \end{bmatrix} \begin{bmatrix} \ddot{c} \\ \ddot{\beta} \end{bmatrix} = \begin{bmatrix} -a\ddot{\alpha}\sin\alpha - a\dot{\alpha}^2\cos\alpha - b\dot{\beta}^2\cos\beta \\ a\ddot{\alpha}\cos\alpha - a\dot{\alpha}^2\sin\alpha - b\dot{\beta}^2\sin\beta \end{bmatrix}$$

$$\begin{cases} \ddot{c} = 0,101 \text{ m/s}^2 = |\vec{a}_B| \\ \ddot{\beta} = 0,696 \text{ rad/s}^2 \end{cases}$$



$$\begin{aligned} \sum F_x = 0 & \quad T + M_{2B} = R_B \cos 15 \\ \sum F_y = 0 & \quad N = M_g + R_B \sin 15 \\ \text{At the bottom of the incline} & \quad T = f_d N = f_d (M_g + R_B \sin 15) \end{aligned}$$

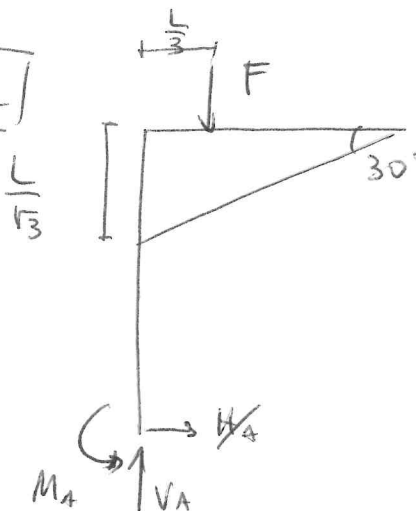
$$\begin{cases} R_B = \frac{M(f_d g + a_B)}{\cos 15 - f_d \sin 15} = 22,57 \text{ N} \\ N = 103,94 \text{ N} \\ T = 20,79 \text{ N} \end{cases}$$

$$\frac{d\vec{r}_C}{dt} = m\vec{v}_A \times \vec{v}_A + J\vec{\omega} \times \vec{\omega} + M\vec{v}_B \times \vec{v}_B = m\frac{a^2}{4}\dot{\alpha}^2 + J\dot{\alpha}^2 + M\dot{c}^2 = 1,61 \text{ W}$$

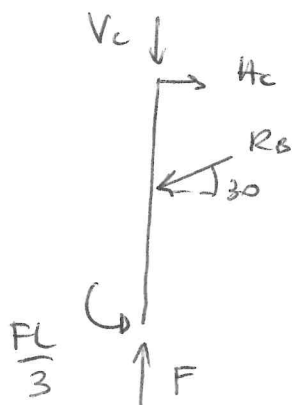
$$\sum W_{\text{ext}} = m\vec{g} \times \vec{v}_A + \vec{F} \times \vec{v}_B - f_d |\vec{N} \vec{v}_B| = -m g \frac{a}{2} \dot{\alpha} \cos \alpha - F \frac{a}{2} \dot{\alpha} \cos 45 - f_d N \dot{c}$$

$$F = \frac{\frac{d\vec{r}_C}{dt} + m g \frac{a}{2} \dot{\alpha} \cos \alpha + f_d N \dot{c}}{-\frac{a}{2} \dot{\alpha} \cos 45} = 37,65 \text{ N}$$

ES 2



$$\begin{cases} \sum F_x = 0 \\ \sum F_y = 0 \\ \sum M_A = 0 \end{cases} \Rightarrow \begin{cases} H_A = 0 \\ V_A = F \\ M_A = \frac{FL}{3} \end{cases}$$

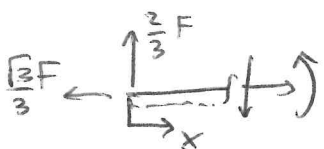
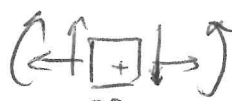
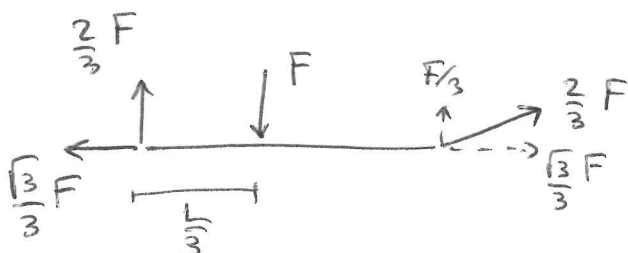


$$\begin{cases} \sum F_x = 0 \\ \sum F_y = 0 \\ \sum M_b = 0 \end{cases} \Rightarrow \begin{cases} H_c = R_b \cos 30 = R_b \frac{\sqrt{3}}{2} \\ F = V_c + R_b \sin 30 = V_c + \frac{R_b}{2} \\ H_c \frac{L}{\sqrt{3}} = \frac{FL}{3} \end{cases}$$

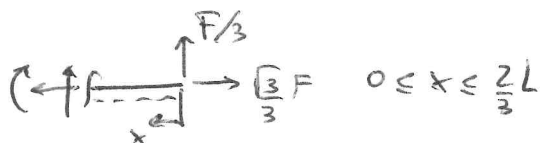
$$\begin{aligned} \rightarrow H_c &= \frac{\sqrt{3}}{3} F \\ \rightarrow R_b &= \frac{2}{\sqrt{3}} H_c = \frac{2}{3} F \\ \rightarrow V_c &= F - \frac{R_b}{2} = \frac{2}{3} F \end{aligned}$$

Recap:

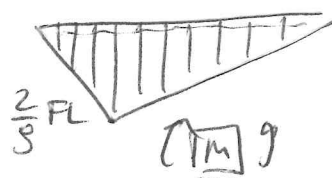
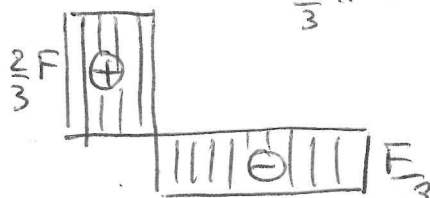
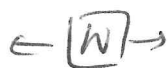
$H_A = 0$	$R_b = \frac{2}{3} F$	$H_c = \frac{\sqrt{3}}{3} F$	$R_d = \frac{2}{3} F$
$V_A = F$		$V_c = \frac{2}{3} F$	
$M_A = \frac{FL}{3}$			



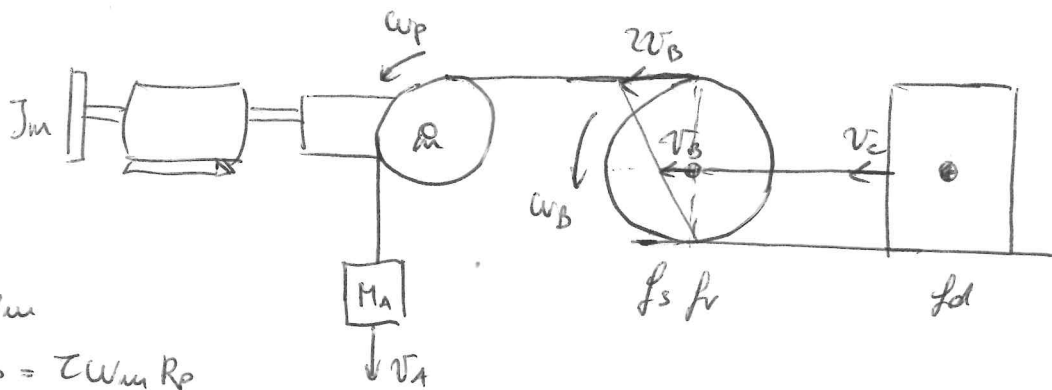
$$0 \leq x \leq \frac{L}{3} \quad \begin{cases} N = \frac{\sqrt{3}}{3} F \\ T = \frac{2}{3} F \\ M = \frac{2}{3} Fx = \begin{cases} 0 & (x=0) \\ \frac{2}{9} FL & (x=\frac{L}{3}) \end{cases} \end{cases}$$



$$0 \leq x \leq \frac{2L}{3} \quad \begin{cases} N = \frac{\sqrt{3}}{3} F \\ T = -\frac{F}{3} \\ M = \frac{F}{3} x = \begin{cases} 0 & (x=0) \\ \frac{2}{9} FL & (x=\frac{L}{3}) \end{cases} \end{cases}$$



ES 3



$$\omega_p = \tau \omega_m$$

$$V_A = \omega_p R_p = \tau \omega_m R_p$$

$$V_B = \frac{V_A}{2} = \tau \omega_m \frac{R_p}{2} = V_C$$

$$\omega_B = \frac{V_B}{R_B} = \tau \frac{\omega_m R_p}{2 R_B}$$

$$W_1 = (C_m - J_m \dot{\omega}_m) \omega_m$$

$$\begin{aligned} \Sigma W_{Att}^{(v)} &= M_A g V_A - f_r R_B N_B \omega_B - f_d N_C V_C = \\ &= M_A g \tau \omega_m R_p - f_r R_B / M_B g \tau \frac{\omega_m R_p}{2 R_B} - f_d M_C g \tau \omega_m \frac{R_p}{2} = \\ &= \left(M_A - f_r \frac{M_B}{2} - f_d \frac{M_C}{2} \right) g \tau R_p \omega_m = C_m^* \omega_m = \underline{\underline{(4,25 \text{ Nm}) \omega_m}} \end{aligned}$$

$$\begin{aligned} \frac{d\vec{L}^{(v)}}{dt} &= M_A V_A a_A + J_p \omega_p \dot{\omega}_p + J_B \omega_B \dot{\omega}_B + M_B V_B a_B + M_C V_C a_C = \\ &= M_A \tau^2 R_p^2 \omega_m \dot{\omega}_m + J_p \tau^2 \omega_m \dot{\omega}_m + J_B \frac{\tau^2}{4} \left(\frac{R_p}{R_B} \right)^2 \omega_m \dot{\omega}_m + (M_B + M_C) \frac{\tau^2}{4} R_p^2 \omega_m \dot{\omega}_m = \\ &= \left[M_A R_p^2 + J_p + \frac{J_B}{4} \left(\frac{R_p}{R_B} \right)^2 + (M_B + M_C) \frac{R_p^2}{4} \right] \tau^2 \omega_m \dot{\omega}_m = \\ &= J_u^* \omega_m \dot{\omega}_m = \underline{\underline{(0,0355 \text{ kgm}^2) \omega_m \dot{\omega}_m}} \end{aligned}$$

① Regime

$$W_2 = \Sigma W_{Att}^{(v)} = C_m^* \omega_m > 0 \rightarrow \underline{\text{Retroposto}}$$

$$W_1 + \eta_R W_2 = 0 \rightarrow C_m \omega_m + \eta_R C_m^* \omega_m = 0 \rightarrow C_m = -\eta_R C_m^* = \underline{\underline{-3,4 \text{ Nm}}}$$

$$\omega_m = \frac{C_{mo} - C_m \omega_s}{C_{mo}} = \underline{\underline{284 \text{ rad/s}}}$$

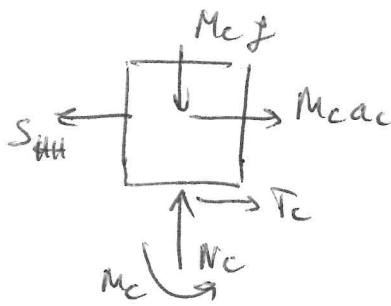
② $a_{mt} = 1,5 \text{ m/s}^2 \rightarrow \dot{\omega}_m = \frac{a_{mt}}{\tau R_p} = 30 \text{ rad/s}^2$

$$W_2 = \Sigma W_{Att}^{(v)} - \frac{d\vec{L}^{(v)}}{dt} = (C_m^* - J_u^* \dot{\omega}_m) \omega_m = (3,17 \text{ Nm}) \omega_m > 0 \rightarrow \underline{\text{Retroposto}}$$

$$W_1 + \eta_R W_2 = 0 \rightarrow (C_m - J_m \dot{\omega}_m) \omega_m + \eta_R (C_m^* - J_u^* \dot{\omega}_m) \omega_m = 0$$

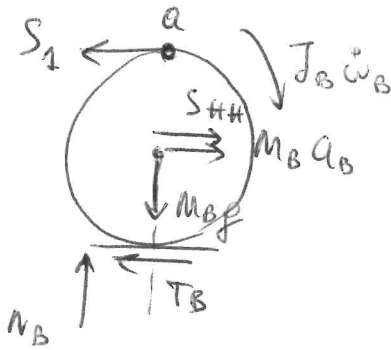
$$C_m = J_m \dot{\omega}_m - \eta_R (C_m^* - J_u^* \dot{\omega}_m) = \underline{\underline{12,46 \text{ Nm}}}$$

③



$$\begin{cases} S_{HH} = M_c a_c + T_c \\ T_c = f_d N = f_d M_c g \\ S_{HH} = M_c (a_c + f_d g) = 28,27 \text{ N} \end{cases}$$

④



$$\sum \Pi_A = 0$$

$$-J_B \ddot{\omega}_B - N_B f_v R_B - T_B 2R_B + (S_{HH} + M_B a_B) R_B = 0$$

$$T_B = \frac{-J_B \ddot{\omega}_B - N_B f_v R_B + (S_{HH} + M_B a_B) R_B}{2R_B} = 16,16 \text{ N}$$

$$T_{lim} = f_s N = f_s M_B g = 54,93 \text{ N}$$

$$T_B < T_{lim} \quad \text{ok} \quad \checkmark$$